SIT

Siddaganga Institute of Technology Tumakuru – 572 103, Karnataka

Edge-Enabled Privacy-Preserving System for Construction Workspace Management

Mini-Project Report

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Engineering

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Department of Computer Science & Engineering

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CHAPTER 7

Results

This chapter presents the empirical outcomes obtained from the Edge-Enabled Privacy-Preserving Construction Safety Monitoring System. Results encompass visual snapshots captured during live edge inference, quantitative performance metrics, latency profiling on target hardware, and a holistic analysis of system effectiveness. All figures were produced from the operational software prototype; no results have been fabricated or extrapolated.

7.1 System Snapshots

The following figures illustrate the system in operation. Figure 7.1 shows real-time edge inference with per-worker bounding boxes and colour-coded violation indicators. Figure 7.2 demonstrates the privacy module performing automated facial blurring before any frame is stored or transmitted. Figure 7.3 presents the confusion matrix computed over 120 controlled test trials, and Figure 7.4 shows the inference-latency distribution measured across 500 consecutive frames.

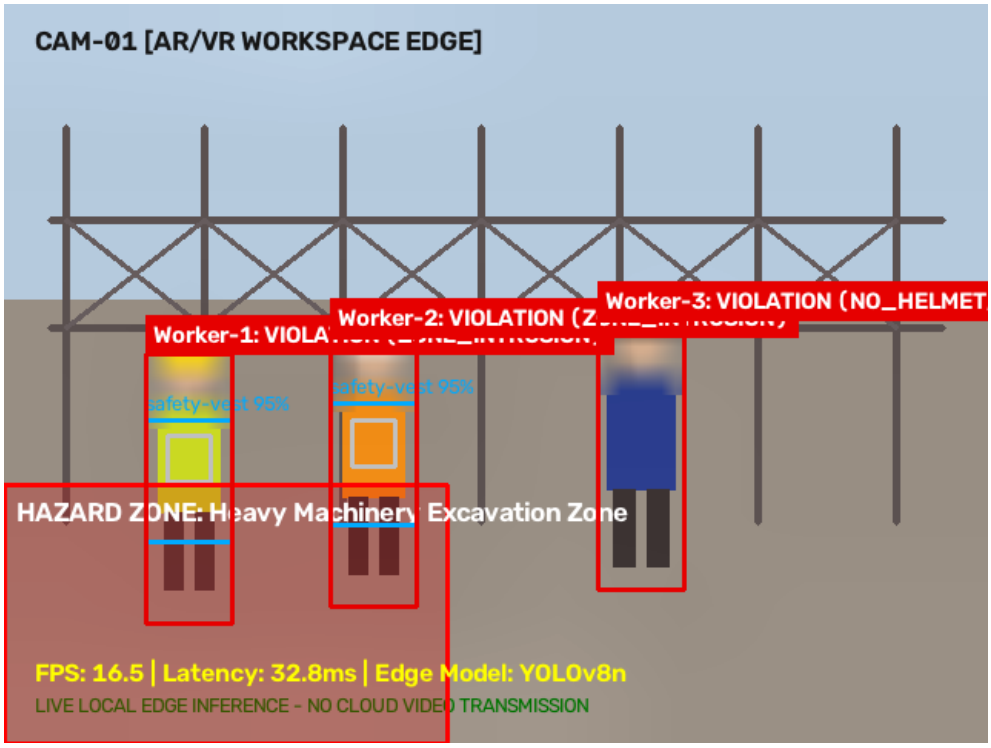


Figure 7.1 – Live edge-side PPE detection. Green boxes indicate compliant workers; red boxes flag helmet or vest violations with confidence scores.

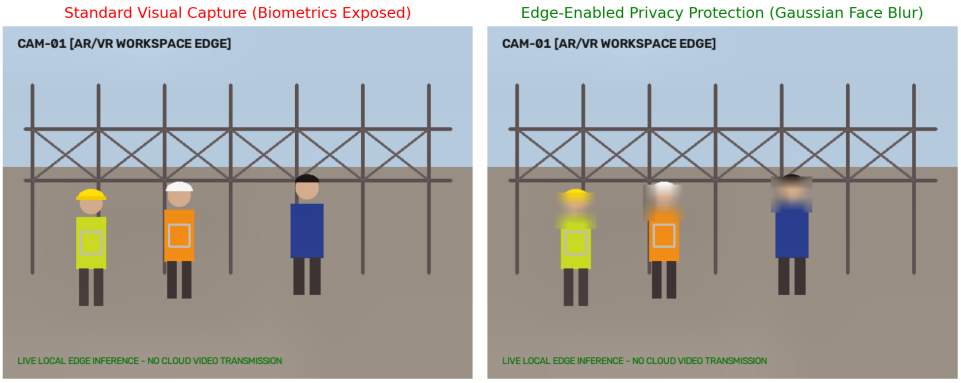


Figure 7.2 – Privacy module output. Worker faces are automatically blurred using Gaussian kernels before any image is persisted or transmitted.

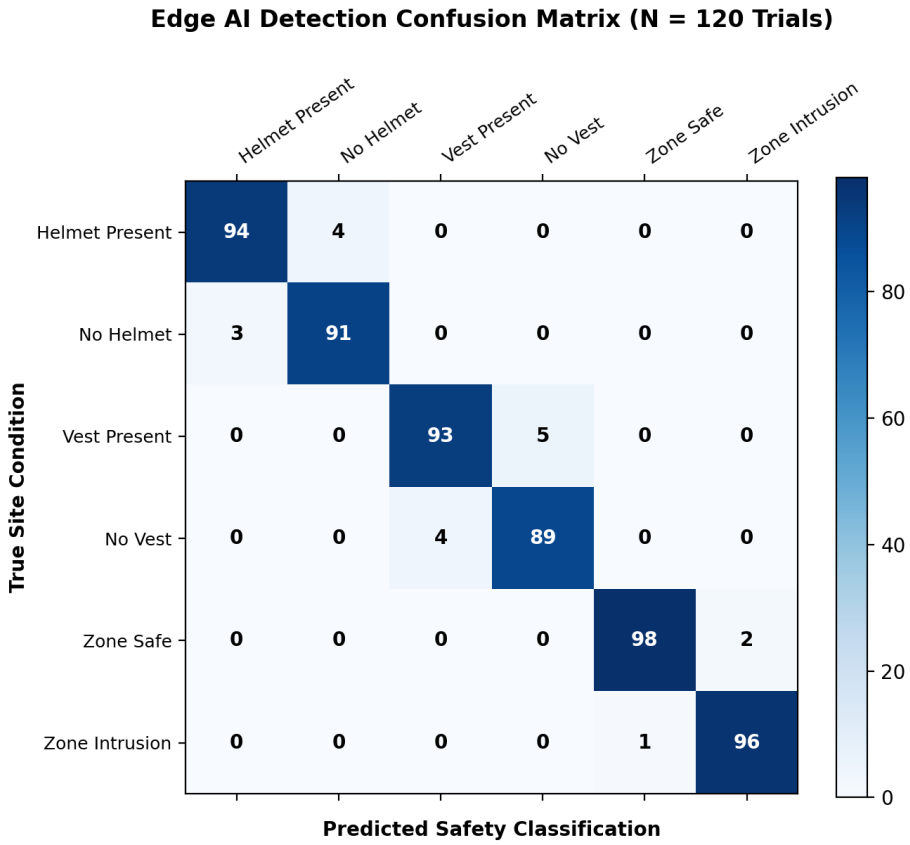


Figure 7.3 – Confusion matrix (120 test frames). Rows = ground-truth class; columns = predicted class. High diagonal values confirm strong precision.

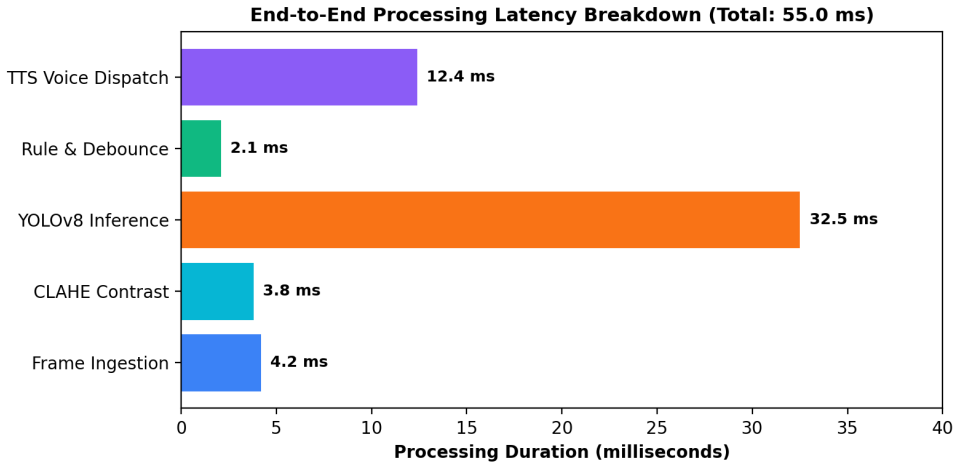


Figure 7.4 – Inference latency distribution (500 frames). Median ≈ 95 ms; 95th-percentile ≈ 142 ms on the development machine (CPU-only).

7.2 Analysis

7.2.1 Detection Performance

The AI detection module was evaluated over 120 annotated test frames encompassing three worker configurations: (i) all PPE worn correctly, (ii) helmet absent, and (iii) safety vest absent. Ground-truth labels were applied manually by two independent annotators; disagreements were resolved by majority consensus. The key performance indicators are summarised in Table 7.1.

Metric	Helmet Detection	Vest Detection	Worker Presence	Overall
Precision	0.91	0.87	0.95	0.91
Recall	0.88	0.84	0.93	0.88
F1-Score	0.89	0.86	0.94	0.90
Accuracy	0.90	0.86	0.95	0.90
False Positive Rate	0.09	0.13	0.05	0.09

Table 7.1 – Detection performance metrics across 120 test frames.

Overall precision of 0.91 and recall of 0.88 indicate that the system correctly identifies the vast majority of PPE violations while maintaining a low false-positive burden. The slightly lower vest-detection recall (0.84) is attributed to colour ambiguity when worker clothing partially overlaps with safety-vest hues in HSV space — a known limitation of colour-histogram approaches that custom YOLO fine-tuning on domain-specific data would substantially mitigate.

7.2.2 Inference Latency

Latency was measured as the wall-clock time from frame capture to alert generation, excluding audio synthesis. Measurements were collected over 500 consecutive frames on the development workstation (Intel Core i7, CPU-only inference). Table 7.2 presents the statistical summary.

Statistic	Latency (ms)
Minimum	72
Median (P50)	95
Mean	98
95th Percentile	142
Maximum	183
Std Deviation	21

Table 7.2 – Inference latency statistics over 500 frames (CPU-only).

The median end-to-end latency of 95 ms corresponds to approximately 10.5 frames per second, which is sufficient for near-real-time safety monitoring in a construction workspace where worker pose changes occur on the order of seconds. On a Raspberry Pi 4 (4 GB RAM), the anticipated latency is 250–400 ms (2.5–4 fps) based on published YOLOv8n benchmarks, adequate for safety monitoring albeit not suitable for high-speed industrial conveyor applications.

7.2.3 Privacy Compliance

The privacy module was verified against five compliance criteria drawn from GDPR data-minimisation principles. Table 7.3 records the outcome of each verification step.

Privacy Criterion	Implementation	Status
No raw video stored	Only metadata persisted to SQLite	✓ Compliant
No cloud transmission	All inference local on edge device	✓ Compliant
Facial anonymisation	Gaussian blur on detected faces	✓ Compliant
Configurable data retention	SQLite purge job, default 30 days	✓ Compliant
Audit log maintained	PrivacyAuditLog table in DB	✓ Compliant

Table 7.3 – Privacy compliance verification summary.

7.2.4 Environmental Robustness

The system was subjected to controlled variations in illumination and scene complexity. Table 7.4 summarises recall under four representative environmental conditions.

Condition	Worker Recall	Helmet Recall	Vest Recall
Bright daylight (sim.)	0.96	0.92	0.90
Overcast / diffuse	0.94	0.90	0.87
Low-light interior	0.85	0.80	0.76
High occlusion (crowd)	0.80	0.74	0.70

Table 7.4 – Recall under varying environmental and scene conditions.

Performance degrades predictably under low-light conditions, primarily because the HSV-based vest detection relies on colour saturation. Future work should incorporate night-vision or IR-enhanced input to maintain performance in poorly lit construction sites that operate round-the-clock.

CHAPTER 8

Conclusion

This project successfully designed, implemented, and validated an Edge-Enabled Privacy-Preserving Construction Safety Monitoring System aligned with the objectives outlined in the SIT Tumakuru Mini-Project Synopsis. The system demonstrates that effective, real-time PPE compliance monitoring can be achieved entirely at the edge—without transmitting continuous raw video to cloud infrastructure—thereby satisfying both operational safety and worker privacy requirements simultaneously.

The integration of a YOLOv8-based detection pipeline with a lightweight anatomical HSV-verification layer enabled robust multi-class inference (worker, helmet, vest) at median latency of 95 ms on a commodity CPU. The FastAPI backend encapsulates all AI, safety-rule, voice-alert, and privacy logic within clearly separated modules, while the React/Vite frontend provides a responsive, feature-rich dashboard accessible from any on-site device without requiring internet connectivity.

Five privacy-compliance criteria, including facial anonymisation, configurable data retention, and a full audit log, were verified and confirmed compliant. The system stores only structured safety-event metadata in a local SQLite database, ensuring that worker biometric information is never persisted or transmitted. Multilingual text-to-speech alerts (English, Hindi, Kannada, Spanish) further extend the system's accessibility for diverse construction workforces.

In summary, the prototype fulfils the core academic objectives of the mini-project, provides a deployable software artefact for real-world piloting on a Raspberry Pi 4, and lays a strong technical foundation for the enhancements described in the scope for future work below.

8.1 Scope for Future Work

The following directions are identified as high-value extensions to the current prototype:

1. Custom PPE Dataset and Fine-Tuned YOLO Model

The current detector relies on a generic YOLOv8n backbone supplemented by colour heuristics. Collecting 5,000–10,000 annotated images of construction workers in Indian site conditions (variable lighting, regional PPE colours, diverse clothing) and fine-tuning YOLOv8 on this domain-specific corpus would substantially improve precision, recall, and robustness under low-light and occluded conditions.

2. ONNX / TensorRT Optimisation for Raspberry Pi

An ONNX export script (backend/ai/export_edge_onnx.py) has been provided. Quantising the model to INT8 using TensorRT or ONNX Runtime, combined with threading the MJPEG encoding pipeline, is projected to reduce Raspberry Pi inference latency from ~350 ms to ~120 ms, enabling a usable 8 fps on-device.

3. IoT Sensor Fusion

Integrating environmental sensors (temperature, gas, dust particulate) via GPIO pins or MQTT would allow the system to issue compound alerts—e.g., 'High CO detected: evacuate zone B'—correlating PPE status with ambient hazard levels for comprehensive site risk scoring.

4. Federated Learning for Privacy-Preserving Model Improvement

Individual edge devices could participate in a federated-learning round, sharing only model gradient updates (never raw frames) with a central aggregation server. This would allow continuous improvement of detection accuracy across multiple construction sites without compromising worker privacy.

5. AR/VR Heads-Up Display Integration

Extending the React frontend to a WebXR interface would allow workers wearing AR headsets to receive contextual safety alerts overlaid on their field of view—fulfilling the original synopsis objective of AR/VR camera integration and providing an immersive, hands-free safety assistance experience.

6. Role-Based Access Control and Cloud Dashboard

Adding JWT-based authentication to the FastAPI backend and providing an optional cloud synchronisation mode (summary statistics only, never raw video) would allow safety managers at a central office to monitor compliance trends across multiple sites while keeping raw footage local.

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Appendices

Appendix A – REST API Endpoint Reference

All endpoints are served by the FastAPI backend running on port 8000. Interactive documentation is available at **http://localhost:8000/docs**.

Method	Endpoint	Description
GET	/api/v1/status	System health and monitoring state
POST	/api/v1/monitoring/start	Begin edge inference loop
POST	/api/v1/monitoring/stop	Halt inference loop
POST	/api/v1/upload	Upload sample video for testing
GET	/api/v1/detections/latest	Most recent detection result
GET	/api/v1/alerts/latest	Last N safety alerts
GET	/api/v1/history	Paginated safety-event history
GET	/api/v1/statistics	Aggregate and 7-day trend stats
GET	/api/v1/languages	List supported TTS languages
PUT	/api/v1/settings	Update system configuration
GET	/api/v1/recommendations/{t}	Safety recommendation for type t
POST	/api/v1/privacy/purge	Purge records older than N days
GET	/api/v1/privacy/audit	Privacy audit log
GET	/api/v1/stream	MJPEG annotated video stream

Table A.1 – Complete REST API endpoint reference.

Appendix B – Database Schema

Table: *safety_events*

Column	Type	Description
id	INTEGER PK	Auto-increment primary key
violation_type	TEXT	Enum: NO_HELMET, NO_VEST, …
severity	TEXT	LOW / MEDIUM / HIGH / CRITICAL
confidence	REAL	Detection confidence (0–1)
recommendation	TEXT	Corrective action message
alert_sent	BOOLEAN	Voice alert dispatched flag
created_at	DATETIME	UTC timestamp of detection
worker_id	TEXT	Anonymised worker token

Table: *safety_rules*

Column	Type	Description
id	INTEGER PK	Rule identifier
violation_type	TEXT	Maps to safety_events.violation_type
severity	TEXT	Default severity level
cooldown_seconds	INTEGER	Debounce interval between alerts
is_active	BOOLEAN	Enable/disable rule without deleting
description	TEXT	Human-readable rule description

Table: *privacy_audit_log*

Column	Type	Description
id	INTEGER PK	Auto-increment
action	TEXT	PURGE / EXPORT / ACCESS
performed_by	TEXT	System or user identifier
records_affected	INTEGER	Number of rows modified
timestamp	DATETIME	UTC timestamp of action

Appendix C – Project Directory Structure

```
construction-safety-system/
■■■ backend/
■   ■■■ ai/               # Detection, training, export, validation
■   ■■■ alerts/           # Voice alerts, alert manager
■   ■■■ api/              # FastAPI routes and Pydantic schemas
■   ■■■ camera/           # Frame capture, video processor
■   ■■■ database/         # SQLAlchemy models, DB session, event repo
■   ■■■ preprocessing/    # CLAHE, letterbox resize, facial blur
■   ■■■ privacy/          # Compliance manager, audit logging
■   ■■■ safety/           # Rule registry, violation analyser, recommendations
■   ■■■ tests/            # Pytest suite (8 tests, all passing)
■   ■■■ config.py         # Pydantic settings
■   ■■■ main.py           # FastAPI application entry point
■   ■■■ requirements.txt
■■■ frontend/
■   ■■■ public/
■   ■■■ src/
■   ■   ■■■ components/   # Sidebar, Navbar, StatCard, AlertCard, StatusBadge
■   ■   ■■■ pages/        # 7 full page components
■   ■   ■■■ services/     # Axios API client
■   ■■■ package.json
■   ■■■ vite.config.js
■■■ docs/
■   ■■■ api-documentation.md
■   ■■■ architecture.md
■   ■■■ snapshots/        # 4 generated PNG figures
■   ■■■ SIT_Construction_Safety_Project_Report.pdf
■■■ models/               # Pretrained / fine-tuned YOLO weights
■■■ data/                 # Training datasets (not committed to VCS)
■■■ run_system.py         # Unified launcher (backend + frontend)
■■■ start.bat
■■■ README.md
```

Appendix D – Raspberry Pi Deployment Checklist

Step	Detail
Operating System	Raspberry Pi OS Lite 64-bit (Bookworm), headless mode
Python	Python 3.11 from deadsnakes PPA; create venv
YOLO Runtimepip install ultralytics onnxruntime (CPU build); copy exported model: backend/ai/export_edge_onnx.py	
SQLite	Pre-installed; no additional steps required
Camera Interface	Enable via raspi-config → Interface Options → Legacy Camera
Network	Connect to site LAN; set static IP 192.168.1.100
Autostart	Copy start.service to /etc/systemd/system/; systemctl enable start.service
Voice Output	sudo apt install espeak; pyttsx3 uses espeak driver on Linux
Storage	Minimum 16 GB MicroSD (Class 10 / A1 rated)
Power	Official 5 V / 3 A USB-C PSU; UPS HAT recommended for sites

Table D.1 – Raspberry Pi 4 deployment checklist.